

RAMCO INSTITUTE OF TECHNOLOGY

**DEPARTMENT OF ELECTRICAL AND ELECTRONICS
ENGINEERING**

COURSE OUTCOMES

REGULATION 2017

Course Code & Title : HS 8151 Communicative English	
Year / Semester : I Year/I Semester	
Course Index : C101	
C101.1	Communicate clearly both in the written form and orally using appropriate vocabulary and comprehend written texts to make inferences.
C101.2	Speak persuasively in different social contexts and write biographical details and technical documents cohesively, coherently and flawlessly using appropriate words.
C101.3	Speak, read and write effectively for a variety of professional and social settings.
C101.4	Read descriptive, narrative, expository and interpretive texts and write using creative, critical, analytical and evaluative methods.
C101.5	Listen, comprehend and respond to different spoken and written discourses/excerpts in different accents and write different genres of texts adopting various writing strategies.
Course Code & Title : MA8151 Engineering Mathematics - I	
Year / Semester : I Year/I Semester	
Course Index : C102	
C102.1	Use both the limit definition and rules of differentiation to differentiate functions.
C102.2	Apply differentiation to solve maxima and minima problems.
C102.3	Evaluate integrals both by using Riemann sums and by using the Fundamental Theorem of Calculus, also evaluate integrals using techniques of integration, such as substitution, partial fractions and integration by parts, in addition to determine convergence/divergence of improper integrals and evaluate convergent improper integrals.
C102.4	Apply integration to compute multiple integrals, area, volume, integrals in polar coordinates, in addition to change of order and change of variables.
C102.5	Apply various techniques in solving differential equations.
Course Code & Title : PH8151 Engineering Physics	
Year / Semester : I Year/I Semester	
Course Index : C103	
C103.1	Analyze the elastic nature of materials and be able to choose the materials depending upon the modulus of elasticity for different applications.
C103.2	Illustrate the advantages of optical communication using LASER.
C103.3	Explain the conducting properties of solids, liquids, good thermal conductor and bad thermal conductors
C103.4	Apply the knowledge of quantum mechanics and classical mechanics in addressing the problems related to science and technology
C103.5	Describe the crystal structures, crystal defects and various crystal growth techniques.
Course Code & Title : CY 8151 & Engineering Chemistry	
Year / Semester : I Year/I Semester	
Course Index : C104	

C104.1	Describe the importance of water technology in the purification of water and its domestic and industrial applications.
C104.2	Illustrate the concepts of adsorption in surface chemistry and catalysis and its applications.
C104.3	Review the use of phase rule in identifying its applications in metallurgy and manufacture of alloys.
C104.4	Compare the different types of industrial techniques of petroleum processing and the determination of calorific values and combustion parameters.
C104.5	Explain the fundamentals of different alternative sources of energy, the generation processes and batteries.
Course Code & Title : GE8151 Problem Solving and Python Programming Year / Semester : I Year/I Semester Course Index : C105	
C105.1	Develop algorithmic solutions to simple computational problems.
C105.2	Read, write and execute simple python programs.
C105.3	Apply control, looping structures and functions to solve problems.
C105.4	Represent compound data using python lists, tuples, and dictionaries.
C105.5	Read and Write data from/to files in python programs.
Course Code & Title : GE8152 Engineering Graphics Year / Semester : I Year/I Semester Course Index : C106	
C106.1	Familiarize with the fundamentals and standards of Engineering graphics
C106.2	Project orthographic projections of lines and plane surfaces
C106.3	Draw the projections of a simple solids and truncated solids
C106.4	Draw the sectional view and develop the lateral surface of simple and truncated solids.
C106.5	Visualize and to project isometric and perspective sections of simple solids.
Course Code & Title : GE8161 Problem Solving and Python Programming Laboratory Year / Semester : I Year/I Semester Course Index : C107	
C107.1	Write, test, and debug simple Python programs.
C107.2	Implement Python programs with conditionals and loops.
C107.3	Develop Python programs step-wise by defining functions and calling them.
C107.4	Demonstrate the use Python lists, tuples, and dictionaries for representing compound data.
C107.5	Illustrate the concepts of read and write data from/to files in Python.
Course Code & Title : BS8161 Physics and Chemistry Laboratory Year / Semester : I Year/I Semester Course Index : C108	
C108.1	Test materials by using their knowledge of applied physics principles in optics and properties of matter.
C108.2	Gain hands -on knowledge in the quantitative chemical analysis of chloride, dissolved oxygen, hardness, alkalinity and copper ions by titration methods.

C108.3	Understand basic concepts in the determination of acids, sodium, potassium and iron by the instrumental methods of analysis.
Course Code & Title : HS 8251 Technical English Year / Semester : I Year/II Semester Course Index : C109	
C109.1	Read technical texts and write area specific texts effortlessly.
C109.2	Listen and comprehend lectures and talks in their areas of specialization and write effectively for a variety of professional and social settings.
C109.3	Speak and write appropriately and effectively in varied formal and informal contexts.
C109.4	Write effectively and persuasively and produce different types of writing such as letters, minutes, reports and winning job applications.
C109.5	Communicate clearly using technical vocabulary in their professional correspondences.
Course Code & Title : MA8251 Engineering Mathematics - II Year / Semester : I Year/II Semester Course Index : C110	
C110.1	Students will have a good understanding of Eigenvalues and eigenvectors, diagonalization of a matrix, Symmetric matrices, Positive definite matrices and similar matrices.
C110.2	Students will have a good understanding of Gradient, divergence and curl of a vector point function and related identities, Evaluation of line, surface and volume integrals using Gauss, Stokes and Green's theorems and their verification.
C110.3	Students will have a good understanding of Analytic functions and conformal mapping.
C110.4	Students will have a good understanding of Complex integration.
C110.5	Students will have a good understanding of Laplace transform and inverse transform of simple functions, properties, various related theorems and application to differential equations with constant coefficients.
Course Code & Title : PH8253 Physics for Electronics Engineering Year / Semester : I Year/II Semester Course Index : C111	
C111.1	Explain the properties of conducting materials using classical and quantum concepts.
C111.2	Apply the fundamental knowledge about the semiconductors and able to differentiate different types of semiconductors.
C111.3	Explain the properties of Magnetic, Dielectric materials and devices for modern day to day applications.
C111.4	Explain the properties and applications of Optical materials and devices.
C111.5	Apply the knowledge about the nano-electronic materials and devices for various applications.
Course Code & Title : BE8252 Basic Civil and Mechanical Engineering	

Year / Semester : I Year/II Semester	
Course Index : C112	
C112.1	Discuss the overview of civil and Mechanical Engineering domain.
C112.2	Explain the usage of construction material and proper selection of construction materials.
C112.3	Identify the building components and structures
C112.4	Identify the components used in power plant cycle and demonstrate working of engines, turbines and pumps
C112.5	Elaborate the components of refrigeration and Air conditioning systems
Course Code & Title : EE8251 Circuit Theory	
Year / Semester : I Year/II Semester	
Course Index : C113	
C113.1	Analyse the circuits using ohm's law, Kirchhoff's laws, mesh current and Node Voltage method
C113.2	Apply the Network Theorems to the analysis of AC and DC circuits.
C113.3	Solve first and second order AC and DC Circuits for Steady – State and transient response in the time domain.
C113.4	Estimate the power consumed by the three- phase balanced circuits.
C113.5	Evaluate the response of the series and parallel resonance circuits, coupled circuits and tuned circuits.
Course Code & Title : GE8291 Environmental Science and Engineering	
Year / Semester : I Year/II Semester	
Course Index : C114	
C114.1	Summarize the importance of environment, biodiversity, ecosystem and how to solve environmental related problems.
C114.2	Describe the causes, effect and control measures of air pollution, water pollution, soil pollution, noise pollution, radioactive pollution and thermal pollution with their relevant case studies.
C114.3	Discuss the various renewable and non-renewable resources and energy conservation processes.
C114.4	Explain the social issues and solutions for sustainable environment with relevant Acts and case studies.
C114.5	Review the impact of human population in the environment and its remedial measures.
Course Code & Title : GE8261 Engineering Practices Laboratory	
Year / Semester : I Year/II Semester	
Course Index : C115	
C115.1	Construct carpentry components and pipe connections including plumbing works.
C115.2	Use welding equipment's to join the structures
C115.3	Illustrate the basic machining operations
C115.4	Construct the models using sheet metal works
C115.5	Describe centrifugal pump, Air conditioner, operations of smithy, foundry and

	fittings.
C115.6	Construct the basic Electrical and Electronics circuits.
C115.7	Examine the different types of electronic circuits and components.
C115.8	Explain the electrical safety rules, grounding, general house wiring.
C115.9	Perform soldering in various electronic circuits.
C115.10	Illustrate the basic operation of domestic electrical appliances.
Course Code & Title : EE8261 Electric Circuits laboratory Year / Semester : I Year/II Semester Course Index : C116	
C116.1	Solve the electrical circuit problems using circuit theorems and laws.
C116.2	Simulate electrical circuits and to experimentally verify various theorems for circuit designing purposes.
C116.3	Experiment the frequency response and transients in passive elements.
C116.4	Simulate the resonance circuits for several applications such as designing of tuning circuit, signal processing and voltage magnification.
C116.5	Perform the simulation of three phase circuits using suitable simulation for both balanced and unbalanced condition.
Course Code & Title : MA8353 Transforms and Partial Differential Equations Year / Semester : II Year/III Semester Course Index : C201	
C201.1	Understand how to solve the given standard partial differential equations.
C201.2	Solve differential equations using Fourier series analysis which plays a vital role in engineering applications.
C201.3	Appreciate the physical significance of Fourier series techniques in solving one and two dimensional heat flow problems and one dimensional wave equations.
C201.4	Understand the mathematical principles on transforms and partial differential equations would provide them the ability to formulate and solve some of the physical problems of engineering.
C201.5	Use the effective mathematical tools for the solutions of difference equations by using Z transform techniques for discrete time systems.
Course Code & Title : EE8391 Digital Logic Circuits Year / Semester : II Year/III Semester Course Index : C202	
C202.1	Solve the problems in number systems and can examine the important characteristics of the digital logic family circuits.
C202.2	Develop the combinational circuits by simplifying the logical expressions using Boolean functions and K maps.
C202.3	Construct various synchronous and asynchronous circuits with its application.
C202.4	Apply the concepts of asynchronous sequential circuits and PLDs.
C202.5	Simulate using software package and development of application oriented logic circuits using VHDL.
Course Code & Title : EE8391 Electromagnetic Theory Year / Semester : II Year/III Semester	

Course Index : C203	
C203.1	Apply vector calculus to static electric-magnetic fields in different engineering situations.
C203.2	Solve electric potential problems in linear, isotropic media for simple geometrics.
C203.3	Determine voltage gradients for simple charge and current configurations and the force between charges and currents.
C203.4	Analyze Maxwell's equation in different forms (differential and integral) and apply them to diverse engineering problems.
C203.5	Examine the phenomena of wave propagation in free space and different media and its interfaces.
Course Code & Title : EE8301 Electrical Machines - I Year / Semester : II Year/III Semester Course Index : C204	
C204.1	Apply the concepts for the study and analysis of magnetic circuits and their applications
C204.2	Explore the operation and performance of single and three phase transformer
C204.3	Describe the electric and magnetic field interactions in electromechanical devices and machines
C204.4	Analyze the working principles, characteristics and performance of DC generator
C204.5	Explain the knowledge on working principle, characteristics, starting, speed control and performance analysis of DC motor
Course Code & Title : EC8353 Electron Devices and Circuits Year / Semester : II Year/III Semester Course Index : C205	
C205.1	Explain the structure and operation of basic electronic devices such as diodes.
C205.2	Illustrate the characteristics of different electronic devices such as transistors and thyristors.
C205.3	Choose and adapt the required components to construct an amplifier circuit.
C205.4	Explore the working of multistage, differential and power amplifiers.
C205.5	Perform design and analysis of feedback amplifiers and oscillators.
Course Code & Title : ME8792 Power Plant Engineering Year / Semester : II Year/III Semester Course Index : C206	
C206.1	Describe the various systems of coal based thermal power plant.
C206.2	Explain the various cycles and systems in diesel, gas and combined cycle power plant.
C206.3	Discuss the working of nuclear reactors in nuclear power plant.
C206.4	Explain the principle/construction and working of hydroelectric and various non-conventional power plants.
C206.5	Analyze energy, economic and environmental issues of power plant.
Course Code & Title : EC8311 Electronics Laboratory	

Year / Semester : II Year/III Semester	
Course Index : C207	
C207.1	Explain the fundamentals of operation and characteristics of semiconductor devices.
C207.2	Calculate the basic parameters of semiconductor devices and their limiting factors.
C207.3	Demonstrate BJT amplifiers in various configurations including DC biasing techniques.
C207.4	Determine the frequency response characteristics of amplifiers.
C207.5	Construct basic electronic circuits, particularly with application to diodes, field-effect transistors and bipolar junction transistors.
Course Code & Title : EE8311 Electrical Machines Laboratory- I	
Year / Semester : II Year/III Semester	
Course Index : C208	
C208.1	Determine the performance of various D.C. Generators and understand the applications of it to power system
C208.2	Troubleshoot the operation of various D.C. Generators
C208.3	Draw the performance characteristics of various D.C. Motors and understand the applications of it to power system
C208.4	Predetermine the performance of DC motor by conducting suitable tests
C208.5	Calculate the voltage regulation and predetermine the performance of the single phase and three phase transformer
Course Code & Title : MA8491 Numerical Methods	
Year / Semester : II Year/IV Semester	
Course Index : C209	
C209.1	Understand the basic concepts and techniques of solving algebraic and transcendental equations.
C209.2	Appreciate the numerical techniques of interpolation and error approximations in various intervals in real life situations.
C209.3	Apply the numerical techniques of differentiation and integration for engineering problems.
C209.4	Expand the given function as Taylor's series and find numerical solutions of ordinary differential equations by different techniques.
C209.5	Determine the numerical solution of one dimensional wave equation and heat equation.
Course Code & Title : EE8401 Electrical Machines -II	
Year / Semester : II Year/IV Semester	
Course Index : C210	
C210.1	Explain the concept, principle and performance of salient and non-salient pole synchronous generator.
C210.2	Describe the concept, principle and performance of synchronous motor.
C210.3	Illustrate the construction, working principle and performance of three phase induction motor.

C210.4	Examine various types of starting and speed control methods of three phase induction motor.
C210.5	Analyze the principle and performance of single phase induction motors and special electrical machines.
Course Code & Title : EE8402 Transmission and Distribution Year / Semester : II Year/IV Semester Course Index : C211	
C211.1	Explain the structure of electric power system and expressions for the computation of transmission line parameters
C211.2	Develop the equivalent circuits for the transmission lines based on distance and to determine voltage regulation and efficiency
C211.3	Analyze the mechanical design of transmission lines and the voltage distribution in insulator strings to improve the efficiency
C211.4	Explain the types, construction of cables and methods to improve the efficiency
C211.5	Explore about distribution systems, types of substations, methods of grounding, EHVAC, HVDC and FACTS
Course Code & Title : EE8403 Measurements and Instrumentation Year / Semester : II Year/IV Semester Course Index : C212	
C212.1	Acquire knowledge on basic functional elements of instruments and various types of errors present in measurements.
C212.2	Explain the various concepts of electrical and electronics instruments.
C212.3	Compare a suitable measuring instrument used for measuring different electrical quantities.
C212.4	Explain the operating principles of various storage and display devices.
C212.5	Explain the operational features of transducer and Data Acquisition System.
Course Code & Title : EE8451 Linear Integrated Circuits and Applications Year / Semester : II Year/IV Semester Course Index : C213	
C213.1	Acquire knowledge in IC fabrication procedure.
C213.2	Determine the DC and AC characteristics of op-amps and signal analysis using op-amp
C213.3	Construct the applications of op-amp based circuits.
C213.4	Examine the operation of special function IC's.
C213.5	Explain the operation of application IC's like voltage regulator and switching regulator.
Course Code & Title : IC8451 Control Systems Year / Semester : II Year/IV Semester Course Index : C214	
C214.1	Develop transfer function models of electrical systems and mechanical systems based on the knowledge of Mathematics, Science and Engineering fundamentals.
C214.2	Derive the time response of various linear systems.
C214.3	Draw the frequency response of the system and examine the effect of compensation on frequency response.

C214.4	Apply the concepts of system stability to analyse performance of closed loop systems.
C214.5	Explain the basic concepts of state variable analysis of systems and calculate the solution of state and output equation.
Course Code & Title : EE8411 Electrical Machines Laboratory-II Year / Semester : II Year/IV Semester Course Index : C215	
C215.1	Determine the regulation of a synchronous generator and explain the applications to power system.
C215.2	Draw the performance characteristics of synchronous motor and explain the applications to power system.
C215.3	Draw the performance characteristics of three phase squirrel cage and slip ring induction motor and understand the applications to power system.
C215.4	Calculate the efficiency and performance characteristics of single phase induction motor.
C215.5	Troubleshoot the operation of various Induction and synchronous machine.
Course Code & Title : EE8461 Linear and Digital Integrated Circuits Laboratory Year / Semester : II Year/IV Semester Course Index : C216	
C216.1	Solve boolean functions and develop adder, subtractor circuits.
C216.2	Perform various code converters to understand the importance of code conversion.
C216.3	Examine the operation of 4 - bit shift registers.
C216.4	Develop Op-Amp in various application circuits
C216.5	Construct counters using specific counter IC.
Course Code & Title : EE8412 Technical Seminar Year / Semester : II Year/IV Semester Course Index : C217	
C217.1	Explain the significance of learning recent advancement in electrical and electronics engineering discipline.
C217.2	Review and prepare the State-of-art technologies in the present technological developments
C217.3	Organize the presentation using the concepts of ordering and determining the central, main and supporting ideas
C217.4	Present any topic in any recent advancement with good communicative skill in front of peers and faculty members
C217.5	Perform well in placement recruitment drive with good technical skills and communication skills
Course Code & Title : EE8501 Power System Analysis Year / Semester : III Year/V Semester Course Index : C301	
C301.1	Classify the power system components such as generator, transformer and transmission lines under steady state operating condition.
C301.2	Solve the power system using iterative techniques for planning, operation and

	control of a power system to calculate the steady state power flow.
C301.3	Determine the short circuit capacity of any power system under symmetrical fault using bus impedance matrix and Thevenin's theorem to calculate fault current.
C301.4	Apply the concept of symmetrical components and sequence networks to analyze the power system when it is subjected to an unsymmetrical fault.
C301.5	Explain the power system under steady state and transient condition and to find the stability of the power system using numerical methods
Course Code & Title : EE8551 Microprocessors and Microcontrollers Year / Semester : III Year/V Semester Course Index : C302	
C302.1	Describe the operation of 8085 microprocessor architecture.
C302.2	Design and implement programs on 8085 microprocessor.
C302.3	Elaborate the operation of 8051 microcontroller architecture.
C302.4	Describe the commonly used peripherals and interfacing.
C302.5	Develop an applications using 8051 microcontroller.
Course Code & Title : EE8552 Power Electronics Year / Semester : III Year/V Semester Course Index : C303	
C303.1	Explain various power semiconductor devices and their switching characteristics.
C303.2	Analyse and choose various AC – DC converters for real time applications.
C303.3	Describe the basic topologies, operation and switching techniques of DC to DC converters.
C303.4	Differentiate different modulation techniques and harmonic reduction methods for inverters.
C303.5	Illustrate the working of AC to AC converters and their applications.
Course Code & Title : EE8591 Digital Signal Processing Year / Semester : III Year/V Semester Course Index : C304	
C304.1	Analyze the various properties of signals and systems, their mathematical representation and the various types of sampling and quantization
C304.2	Analyze discrete time LTI(Linear Time Invariant) systems using Z transform and DTFT(Discrete Time Fourier Transform).
C304.3	Apply the concepts of DFT(Discrete Fourier Transform) and FFT(Fast Fourier Transform) to solve Electrical Engineering Problems.
C304.4	Design Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) digital filters.
C304.5	Explain the architectural features, addressing formats and the various functional modes of DSP Processors.
Course Code & Title : CS8392 Object Oriented Programming Year / Semester : III Year/V Semester Course Index : C305	

C305.1	Develop and implement Java programs for simple applications that make use of classes, packages and inheritance.
C305.2	Develop and implement Java programs with exception handling and multithreading.
C305.3	Develop and implement Java programs using Stack ADT and Array List using Interfaces
C305.4	Design applications using file processing, generic programming and event handling.
C305.5	Develop and deploy mini-projects using Java Concepts.
Course Code & Title : OCE551 Air Pollution and Control Engineering Year / Semester : III Year/V Semester Course Index : C306	
C306.1	Describe the nature and characteristics of air pollutants with its impact on human, vegetation, animals and properties.
C306.2	Review the meteorological effects of air pollution
C306.3	Explain the design stacks and particulate air pollution control devices to meet applicable standards.
C306.4	Identify equipment's to control gaseous contaminants.
C306.5	Summarize the sources, types and control measures of indoor air pollutants and noise pollution.
Course Code & Title : OIT551 Database Management Systems Year / Semester : III Year/V Semester Course Index : C307	
C307.1	Interpret relational data model, evolve conceptual model of a given problem, its mapping to relational model.
C307.2	Query the relational database and write simple and complex queries
C307.3	Develop and write programs with database connectivity.
C307.4	Analyze and design a database system and perform normalization.
C307.5	Identify the concepts of database security and information retrieval systems.
Course Code & Title : OMF551 Product Design and Development Year / Semester : III Year/V Semester Course Index : C308	
C308.1	Describe the concepts of product development Processes and Organizations
C308.2	Outline the principles of opportunity identification, creating, identifying, and screening ideas for new products.
C308.3	Explain the implications of product architecture on product change, product variety, component standardization, product performance, manufacturing cost, and project management.
C308.4	Illustrate the techniques of industrial design of a product through prototyping technology.
C308.5	Discuss various techniques used to reduce manufacturing cost and decrease the environmental impacts associated with products.
Course Code & Title : OAN551 Sensors and Transducers	

Year / Semester : III Year/V Semester	
Course Index : C309	
C309.1	Expertise in various calibration techniques and signal types for sensors
C309.2	Apply the various sensors in the Automotive and Mechatronics applications
C309.3	Explain force, magnetic and heading sensor in Aerospace and Home appliances.
C309.4	Describe the basic principles of various smart sensors.
C309.5	Implement the DAQ systems with different sensors for real time applications.
Course Code & Title : EE8511 Control and Instrumentation Laboratory	
Year / Semester : III Year/V Semester	
Course Index : C310	
C310.1	Explain control theory and apply to electrical engineering problems.
C310.2	Analyze the various types of ADC, DAC converters.
C310.3	Design lag, lead and lag-lead compensators.
C310.4	Describe the basics of bridge networks and signal conditioning circuits.
C310.5	Demonstrate the response and stability of control systems using simulation package.
Course Code & Title : HS8581 Professional Communication	
Year / Semester : III Year/V Semester	
Course Index : C311	
C310.1	Cultivate intercultural communication skills, to guide students in making appropriate and responsible decisions, to develop leadership traits and soft skills and to create a desire to fulfill individual goals and team goals.
C310.2	Help the learners acquire listening and speaking skills through lab based activities, and enable them to introduce themselves and make effective presentations.
C310.3	Guide learners to evaluate their thinking skills, acquire listening and speaking skills and enable them to involve in group participation.
C310.4	Teach various formats of interview, answering techniques, body language and paralinguistic skills.
C310.5	Clarify and prioritize learners' objectives and goals, to contribute and work as a team by creating more leadership opportunities.
Course Code & Title : CS8383 Object Oriented Programming Laboratory	
Year / Semester : III Year/V Semester	
Course Index : C312	
C312.1	Develop Java programs using OOP principles.
C312.2	Develop Java programs with the concepts inheritance and interfaces.
C312.3	Build Java applications using exceptions and I/O streams.
C312.4	Develop Java applications with threads and generics classes.
C312.5	Develop interactive Java programs using swings.
Course Code & Title : EE8601 Solid State Drives	
Year / Semester : III Year/VI Semester	
Course Index : C313	
C313.1	Explain the steady state operation and transient dynamics of a motor load

	system.
C313.2	Examine the operation of converter/chopper fed dc drive.
C313.3	Apply the control methods for the induction motor drives for suitable applications.
C313.4	Analyze the operation and performance of synchronous motor drives.
C313.5	Design the current and speed controllers for a closed loop solid state DC motor drives.
Course Code & Title : EE8602 Protection and Switchgear Year / Semester : III Year/VI Semester Course Index : C314	
C314.1	Analyze the knowledge on different protective schemes in power system
C314.2	Explain various electromagnetic relays and its application
C314.3	Choose the protection scheme for various faults in motor, generator, transformer, bus bar, transmission line
C314.4	Examine various static relays and numerical relays and its application
C314.5	Describe concepts and principle of various circuit breakers
Course Code & Title : EE8691 Embedded Systems Year / Semester : III Year/VI Semester Course Index : C315	
C315.1	Explain the building blocks of embedded systems.
C315.2	Analyse the various communications in processors and input/output interfacing.
C315.3	Apply the embedded development strategies to develop the embedded firmware environment.
C315.4	Describe the various scheduling techniques in Real Time Operating System.
C315.5	Apply the concepts of embedded system in application developments within realistic constraints such as economic, environmental and society.
Course Code & Title : EE8002 Design of Electrical Apparatus Year / Semester : III Year/VI Semester Course Index : C316	
C316.1	Explain basics of design consideration for rotating and static electrical machines and design of field system for its application.
C316.2	Describe the design of single and three phase transformer.
C316.3	Discuss the design of armature and field of DC machines.
C316.4	Design stator and rotor of induction motor.
C316.5	Analyze and design synchronous machines.
Course Code & Title : GE8075 Intellectual Property Rights Year / Semester : III Year/VI Semester Course Index : C317	
C317.1	Describe the concepts of various intellectual property rights
C317.2	Elaborate the practical aspects of registration in India and abroad
C317.3	Explain the implications of agreements and legislations
C317.4	Illustrate the methods used for digital content protection
C317.5	Discuss the legal aspects governing IPR infringement

Course Code & Title : EE8005 Special Electrical Machines	
Year / Semester : III Year/VI Semester	
Course Index : C318	
C318.1	Explain the construction, operation and control methods of stepping motors
C318.2	Describe the operation of various power controllers required for switched reluctance motors
C318.3	Derive the expressions for magnetic circuit analysis, emf and torque equations of permanent magnet brushless DC motors
C318.4	Illustrate the construction, operation and performance characteristics of permanent magnet synchronous motors
C318.5	Choose a special machine for a particular application
Course Code & Title : EE8006 Power Quality	
Year / Semester : III Year/VI Semester	
Course Index : C319	
C319.1	Examine various sources, causes and effects of power quality issues in electrical power systems.
C319.2	Analyse the sources and mitigation techniques against voltage sag and swell in power systems.
C319.3	Explain the various sources for harmonics, harmonic indices and harmonics reduction methods.
C319.4	Explain the principle of operation of shunt and series compensators to mitigate various power quality issues also the methods of voltage regulation and power factor correction.
C319.5	Solve the power quality issues by employing custom power devices and power quality monitoring equipments.
Course Code & Title : EE8661 Power Electronics and Drives Laboratory	
Year / Semester : III Year/VI Semester	
Course Index : C320	
C320.1	Demonstrate the gate pulse generation using various configurations and
C320.2	Explain the characteristics of various switches.
C320.3	Illustrate the operation and performance parameters of DC – AC, DC – DC converter circuits.
C320.4	Interpret the performance of AC – DC, AC – AC converter circuits.
C320.5	Simulate various power electronic converter circuits.
Course Code & Title : EE8681 Microprocessors and Microcontrollers Laboratory	
Year / Semester : III Year/VI Semester	
Course Index : C321	
C321.1	Apply the arithmetic operations, logic operations and sorting using 8085 microprocessor.

C321.2	Write a program for code conversion, finding the maximum and minimum numbers in a series.
C321.3	Analyze the operations of peripheral interfacing with 8085 microprocessor and write program using jubins Simulator.
C321.4	Apply the arithmetic, logic operations and branching operation using 8051 microcontroller.
C321.5	Write a program for peripheral interfacing with 8051 microcontroller.
Course Code & Title : EE8611 Mini Project Year / Semester : III Year/VI Semester Course Index : C322	
C322.1	Apply the fundamental knowledge and skills, which are acquired within the technical area to a given problem as well as summarize list of literature review, analyze previous researcher's work and relate them to the project.
C322.2	Design engineering solutions to complex problems and conduct experiments, as well as analyze and interpret the data.
C322.3	Utilize technology tools for communication, collaboration, decision support and also demonstrate the knowledge, skills and attitudes of a professional engineer.
C322.4	Interact with team members in a professional manner, respecting differences, to ensure a collaborative project environment and also demonstrate a strong working knowledge of ethics.
C322.5	Document and present one's own work for a given target group with good oral and written presentation skills and also recognize the need for life-long learning by undergoing the project work
Course Code & Title : EE8701 High Voltage Engineering Year / Semester : IV Year/ VII Semester Course Index : C401	
C401.1	Identify the source, effects and protection methods of over voltages in power system
C401.2	Describe the breakdown mechanism in different dielectrics
C401.3	Explain the different methods of overvoltage generation
C401.4	Explore the various overvoltage measurement methods
C401.5	Apply the concepts of high voltage testing principles to power apparatus
Course Code & Title : EE8702 Power System Operation and Control Year / Semester : IV Year/ VII Semester Course Index : C402	
C402.1	Illustrate the basic concepts of power system operation and control.
C402.2	Construct the power-frequency controller to single area and two area systems.
C402.3	Employ the suitable control actions to maintain the voltage profile against various loads.
C402.4	Schedule the generators in power system economically by unit commitment economic dispatch.
C402.5	Apply the concept of computer control of power system.
Course Code & Title : EE8703 Renewable Energy Systems	

Year / Semester : IV Year/ VII Semester	
Course Index : C403	
C403.1	Explain the importance and limitations of renewable energies using present Indian and International energy scenario.
C403.2	Describe the working of different types of wind power plants and its grid integration issues.
C403.3	Discuss the solar energy harnessing methods along with types, characteristics and applications.
C403.4	Analyze the energy conversion process and the environmental effects on biomass energy, geo thermal energy and hydro power generating power plants.
C403.5	Examine the working of several renewable energy systems such as tidal energy, ocean thermal energy, hydrogen production and storage, Energy storage systems and hybrid systems.
Course Code & Title : OCS752 Introduction to C Programming	
Year / Semester : IV Year/ VII Semester	
Course Index : C404	
C404.1	Implement C programs using basic programming constructs.
C404.2	Develop simple applications using arrays.
C404.3	Design and implement Programs using strings.
C404.4	Develop simple applications using functions.
C404.5	Implement C program using structures..
Course Code & Title : OML751 Testing of Materials	
Year / Semester : IV Year/ VII Semester	
Course Index : C405	
C405.1	Learn about different type of materials, classification and Testing Standards.
C405.2	Analyze engineering components using various mechanical testing procedure.
C405.3	Explain about Principles, Techniques, Advantages and Limitations, Applications of Non-Destructive Testing.
C405.4	Exploit Principles, Types, Advantages and Limitations, Applications of materials characterization process.
C405.5	Ability to use the different technique and know its applications and limitations.
Course Code & Title : EI8075 Fibre Optics and Laser Instrumentation	
Year / Semester : IV Year/ VII Semester	
Course Index : C406	
C406.1	Explain the principle, transmission, dispersion and attenuation characteristics of optical fibers.
C406.2	Describe the selection of the appropriate optical fiber sensors for industrial application.
C406.3	Classify the different types of lasers and discuss their principle, fundamental characteristics and properties.
C406.4	Apply laser theory for the selection of lasers for specific Industrial applications.
C406.5	Discuss the principle of holography and the use of lasers in Medical applications.

Course Code & Title : GE8077 Total Quality Management	
Year / Semester : IV Year/ VII Semester	
Course Index : C407	
C407.1	Discuss the contributions of Quality Guru.
C407.2	Explain the principles of TQM.
C407.3	Apply the tools and techniques of quality management to manufacturing and service processes.
C407.4	Describe TQM tools and techniques such as Cost of Quality, QFD and TPM.
C407.5	Discuss the elements of Quality system standards.
Course Code & Title : EE8711 Power System Simulation Laboratory	
Year / Semester : IV Year/ VII Semester	
Course Index : C408	
C408.1	Develop a suitable program for determining the transmission line parameters and its performance evaluation.
C408.2	Simulate the power system to obtain the power flow solution using numerical iterative methods.
C408.3	Perform fault analysis for a given power system under symmetrical and unsymmetrical fault.
C408.4	Analyze the stability of the power system by simulation using appropriate techniques
C408.5	Develop a suitable program to solve economic dispatch problem and load frequency dynamics of interconnected power system.
Course Code & Title : EE8712 Renewable Energy Systems Laboratory	
Year / Semester : IV Year/ VII Semester	
Course Index : C409	
C409.1	Demonstrate the concept of electric power generation using several renewable energy resources.
C409.2	Analyze the characteristics and shadowing effect of solar PV energy system.
C409.3	Evaluate the performance of Grid connected and Standalone PV system, micro wind generation, fuel cell and Hybrid systems.
C409.4	Simulate various renewable energy technologies such as solar PV system, wind, fuel cell and hybrid energy systems.
C409.5	Analyze the significance of intelligent controllers for hybrid energy generation systems.
Course Code & Title : GE8076 Professional Ethics in Engineering	
Year / Semester : IV Year/ VIII Semester	
Course Index : C410	
C410.1	Apply ethics, morals and human values in society.
C410.2	Explain about engineering ethics.
C410.3	Realize the responsibilities of engineers as experimenters.

C410.4	Recognize the safety, risks, risk benefit analysis and rights of an engineer.
C410.5	Discuss the importance of the global issues, moral leadership and code of conduct.
Course Code & Title : MG8591 Principles of Management Year / Semester : IV Year/ VIII Semester Course Index : C411	
C411.1	Discuss the evolution of management, functions and roles of managers.
C411.2	Explain the different types of planning process and tools used for planning.
C411.3	Elaborate different organization structures and functions of human resources manager.
C411.4	Illustrate the different theories of motivation and leadership.
C411.5	Describe the control techniques and the role of technology in management.
Course Code & Title : EE8019 Smart Grid Year / Semester : IV Year/ VIII Semester Course Index : C412	
C412.1	Describe the basic concepts of smart grid and latest developments
C412.2	Explain various aspects of the smart grid such as technologies, components, architectures and applications
C412.3	Explain about the Advanced Metering Infrastructure and its protocols
C412.4	Apply the knowledge of power quality management in Smart Grids
C412.5	Develop more understanding on LAN, WAN and Cloud Computing for Smart Grid applications
Course Code & Title : EE8811 Project Work Year / Semester : IV Year/ VIII Semester Course Index : C413	
C413.1	The students will be able to apply the fundamental knowledge and skills, which are acquired within the technical area, to a given problem
C413.2	- The students will be able to identify and summarize an appropriate list of literature review, analyze previous researchers' work and relate them to the project - Within given constraints, even with limited information, the students will be able to independently analyze and discuss complex inquiries/problems and handle larger problems on the advanced level within the technical area.
C413.3	- The students will be able to design engineering solutions to complex problems in a systematic approach. - The students will be able to identify and apply appropriate parameters, assumptions and design criteria in consideration of health and safety (example: the use of codes of practice), ethics, economics, environment, sustainability
C413.4	The students will be able to apply research and conduct experiments, as well as to analyze and interpret data that yield the results and answer important applicable research questions.
C413.5	The students will be able to utilize technology tools for communication, collaboration, information management, and decision support.

C413.6	The students will be able to demonstrate the knowledge, skills and attitudes of a professional engineer.
C413.7	The students will be able to interact with team members in a professional manner, respecting differences, to ensure a collaborative project environment.
C413.8	The students will be able to demonstrate a strong working knowledge of ethics and professional responsibility.
C413.9	The students will be able to document and present one's own work, for a given target group, with strict requirements on structure, format, and language usage.
C413.10	The students will be able to present the project outlining the approach and expected results using good oral and written presentation skills.
C413.11	The students will be able to demonstrate effective organizational leadership and change skills for managing projects and project teams.
C413.12	The students will be able recognize the need for life-long learning by undergoing the project work.